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JANUARY 2005

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 03 – General Proficiency

1 hour

14 JANUARY 2005 (p.m.)

In addition to the 1 hour, candidates are allowed a reading time of 10 minutes. Writing may begin during the 10-minute period.

This paper is divided into THREE sections. Candidates MUST answer ONE question from EACH section. Where appropriate, answers should be illustrated by diagrams.

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SECTION A

Answer ONE question from this section.

1. (a) (i) Give TWO differences between osmosis and diffusion. **(2 marks)**

(ii) Describe how osmosis is used in transport by a plant. **(4 marks)**

(b) (i) Explain why controlling osmosis in cells is an important part of the control of diabetes. **(4 marks)**

(ii) Explain why diabetes is considered a physiological disorder and not a nutritional disease, although it can be controlled by changes in the diet. **(2 marks)**

(c) (i) Patients whose kidneys have stopped working efficiently are put on strict diets. They are allowed to have

(i) little salt

(ii) very small quantities of meat, beans and peas

(iii) a few ounces of water when they are thirsty.

Explain why EACH of these adjustments to the diet is necessary. **(4 marks)**

(ii) When using an artificial kidney, blood from the patient passes through tubes made of a selectively permeable membrane. The tubes are surrounded by a watery fluid containing carefully controlled amounts of certain solutes.

Suggest how a kidney machine is able to 'clean' the patient's blood. **(4 marks)**

Total 20 marks

2. (a) (i) Compare the structure of a red blood cell and a white blood cell. **(2 marks)**

(ii) Describe the role of white blood cells in humans. **(4 marks)**

(b) (i) Describe ONE way in which the skin of humans offers protection. **(2 marks)**

(ii) How do the skin and the blood work together in order to continue protecting the body when it is cut? **(4 marks)**

(c) (i) Identify TWO differences between the structure of the epidermal tissue of a leaf and the human skin. **(4 marks)**

(ii) Some plant diseases are caused by bacteria or viruses. These diseases spread rapidly through the plant, sometimes killing it. Explain how the transport system of the plant may be involved. **(4 marks)**

Total 20 marks

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SECTION B

Answer ONE question from this section.

3. (a) (i) Using appropriate examples, distinguish between the physical and the biotic factors of a natural habitat. **(4 marks)**

(ii) How might a biotic factor affect ONE of the physical factors in the habitat? **(2 marks)**

(b) A fire in part of a forest killed most plants, except a few trees. After some weeks plants started to grow again but no animals were found. A year later the community was almost the same as before the fire.

(i) Give TWO methods of dispersal by which some of the new plants may have reached the area.

(ii) Explain why plants must return before the animals can come back. **(4 marks)**

(c) In another part of the same forest mentioned at (b) above, chemical waste was dumped from a nearby factory. All the plants and animals died. After a year nothing had returned.

(i) Suggest why the plants and animals did NOT return after this event. **(4 marks)**

(ii) Discuss TWO long-term effects that could be caused by the permanent loss of a large part of a forest AND suggest how this could be avoided. **(6 marks)**

Total 20 marks

4. (a) (i) Few people have hereditary diseases compared to pathogenic diseases. Give TWO characteristics of these two types of diseases that could account for this. **(2 marks)**

(ii) Using ONE named disease as an example, explain how a baby could have a hereditary disease which neither parent has. **(4 marks)**

(b) (i) What characteristics of AIDS cause it to be considered a deadly disease although people rarely die directly from it? **(4 marks)**

(ii) Do you think it is possible for humans to eventually become resistant to AIDS? Give ONE reason for your answer. **(2 marks)**

(c) Plants brought into a country without permission are taken away from passengers and, in most cases, destroyed. In some countries, persons with AIDS are permanently isolated from other citizens. Many people object to these two practices.

To what extent do you consider these TWO practices acceptable? Support your answer with reasons. **(8 marks)**

Total 20 marks

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SECTION C

Answer ONE question from this section.

5. (a) Your breakfast consists of bread, eggs and butter.

(i) What are the end products of digestion of these foods?

(ii) Describe FULLY what happens in the body to EACH end product after it is digested.

(8 marks)

(b) A student attempted to investigate the actions of the enzyme lipase. He placed some pieces of fat in a test tube of water and added some lipase solution. He left the test tube in an air-conditioned laboratory overnight. The next day there was little change in the fat. Explain FULLY why he did NOT get the results he expected.

(6 marks)

(c) In recent times, people have been cautioned to reduce the amount of fat in their diets.

(i) Why is the reduction of the amount of fat in the diet recommended?

(4 marks)

(ii) Suggest TWO reasons why fat should NOT be completely eliminated from the diet. **(2 marks)**

Total 20 marks

6. (a) (i) Identify the TWO systems in the human body that are responsible for co-ordination.

(ii) Give THREE reasons why co-ordination is important for all living things. Explain your answer.

(8 marks)

(b) During a medical examination of a seven-year-old girl, it is determined that she will be unable to produce adequate quantities of oestrogen. Discuss how this condition may affect her future life. **(6 marks)**

(c) (i) Which hormone progesterone or oestrogen, is present in **greater** quantities in birth contol pills? Explain your answer. **(3 marks)**

(ii) It is possible to remove an ovum from one woman, fertilise it in a test tube and implant the embryo in the uterus of another woman. Suggest THREE problems which may be associated with this procedure. **(3 marks)**

Total 20 marks

END OF TEST